



Elections

Task A: Understanding First Past the Post (FPTP)

1) 200 people are voting in an election. There are four candidates (A, B, C, D) standing in the election.

a) Given that candidate B won the election, suggest a possible number of votes that each candidate received.

There are many possibilities however your solution should have candidate B with the highest number of votes and the number of votes cast should sum to 200.

b) Given that candidate B won the election, what is the smallest number of votes candidate B could have received?

51 votes.

(The other candidates could have received 50, 50 and 49 votes.)

2) 400 people are voting in an election. There are four candidates (A, B, C, D) standing in the election.

a) Given that candidate D won the election, suggest a possible number of votes that each candidate received.

There are many possibilities however your solution should have candidate D with the highest number of votes and the number of votes cast should sum to 400.

b) Given that candidate D won the election, what is the smallest number of votes candidate D could have received?

101 votes.

(The other candidates could have received 100, 100 and 99 votes.)

3) n (where n is a multiple of four) people are voting in an election. There are four candidates (A, B, C, D) standing in the election.

a) Given that candidate A won the election, write a general rule for the smallest number of votes candidate A could have received.

$$\frac{n}{4} + 1$$

b) Write an expression for the proportion of the total vote that candidate A received.

$$\frac{\frac{n}{4} + 1}{n}$$



Task B: Limitations of FPTP

1) Constituency 1 - this is the breakdown of the vote.

Party A	Party B	Party C	Party D
5772	6889	3424	789

a) Which party won the election in this constituency?

Party B as they received more votes than any other party.

b) What percentage of the total vote did each party receive?

(Round to the nearest percent.)

Party A	Party B	Party C	Party D
34%	41%	20%	5%

2) Constituency 2 - this is the breakdown of the vote.

Party A	Party B	Party C	Party D
4930	5381	4729	1291

a) Which party won the election in this constituency?

Party B as they received more votes than any other party.

b) What percentage of the total vote did each party receive?

(Round to the nearest percent.)

Party A	Party B	Party C	Party D
30%	33%	29%	8%

3) Constituency 3 - this is the breakdown of the vote.

Party A	Party B	Party C	Party D
21 953	7116	4581	2093

a) Which party won the election in this constituency?

Party A as they received more votes than any other party.

b) What percentage of the total vote did each party receive?

(Round to the nearest percent.)

Party A	Party B	Party C	Party D
61%	20%	13%	6%



4) By considering your answers to questions 1-3, answer the following:

a) Which party won the election in the most constituencies?

Party B as they won two out of three constituencies.

b) By considering the total number of votes cast across the three constituencies, what percentage of the vote did each party win? Write a paragraph explaining what this means.

68 948 votes cast in total. (Approximate percentages below.)

Party A	Party B	Party C	Party D
47%	28%	18%	6%

Despite Party A receiving the most votes across the constituencies, Party B still won more individual constituencies. Party A may be more popular but it does not have as many elected members as Party B does.